GMI German-Malaysian Institute

DIPLOMA PROGRAMME

INDIVIDUAL TASK

Academic Period: July 2026

Program	:	**CYBER SECURITY TECHNOLOGY (CBS)**
Course	:	**NETWORK MANAGEMENT & MONITORING SYSTEMS**
Code	:	**NTC 1052**

Title	:	**SIMPLE NETWORK MANAGEMENT PROTOCOL**

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Semester/Trade/Group	:	SEMESTER 5 / DCBS / CBS 1
Date	:	11/8/2026

Part 1: Short Answer Questions

1. **Structure of Management Information (SMI)**
    - Explain what the **Structure of Management Information (SMI)** is in network management.
    - **The MIB is an essential part of the simple network management model as it stores and defines the information exchanged within a Simple Network Management Protocol system. It stores collected data for fault management, performance management, and capacity planning.**

    - List two primary data types defined by SMI and briefly explain their purpose.

    - **INTEGER & OCTET STRING**
    - **Integer are used to report numeric measurements, while Octet Strings are used to store and transit text strings.**

2. **Management Information Base (MIB)**
    - Define what a **Management Information Base (MIB)** is and describe its tree-like hierarchical structure.
    - **Text file describing managed objects hierarchically via unique OIDs.**
    - **The database is conceptually organized as a tree. The upper structure of this tree is defined in Requests for Comments (RFC) 1155 and RFC 1213.**

    - Differentiate between standard MIBs and custom (vendor-specific) MIBs.

    - **Standards organizations define common standard MIBs and are widely supported across devices. MIB-II is one of the most common MIBs, providing general network statistics such as interface status, IP addresses, and error counts. While device manufacturers create vendor-specific private MIBs to provide information about proprietary features not covered by standard MIBs. For example, Cisco, Juniper, and HP devices may include private MIBs for advanced routing, firewall functions, or special hardware metrics.**

Part 2: Comparison Table

Fill out the missing details in the table below to compare **SNMPv1**, **SNMPv2c**, and **SNMPv3**:

Feature / Metric	SNMPv1	SNMPv2c	SNMPv3
Primary Security Mechanism	Community Strings (Plaintext)	**Community Strings (Plaintext)**	**User-Based Security (USM) View-based access control (VACM)**
Supported Operations	Get, GetNext, Set, Trap	Adds: *GetBulk*, *InformRequest*	**Adds : Report, SNMPv2-Trap**
Performance & Efficiency	Low (Single-variable retrieval)	**High (Non-atomic execution)**	High
Authentication & Encryption	None	**None**	**HMAC-SHA / MD5 & AES / DES Encryption**

Part 3: Reflection

A network administrator needs to manage router configurations securely over a public internet connection.

- Which SNMP version should they implement?
  SNMPv3
- Provide **two key reasons** why this version is the safest choice for this scenario compared to earlier versions.

  1. **Prevents eavesdropping by protocol analyzers by using encryption. Such as DES, 3DES or AES, to encrypt the payload of the SNMP message.**
  2. **Verifies operator authorization and protects critical data from intentional and/or accidental corruption by using an access control table (part of policy-based management).**

References

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